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Chapter 1

Introduction

High-performance computing (HPC) has become essential to modern science, and its capability has grown by drawing more power, packing more hardware, and cycling through that hardware ever faster (Portegies Zwart, 2020; Allen, 2022). The environmental cost of this growth has traditionally been equated with operational energy consumption (Koomey *et al.*, 2011), yet operational energy is only one facet of the footprint. Manufacturing the processors, memory, and interconnects that populate a machine emits carbon long before the system is switched on, and this embodied share is rising as fabrication grows more complex (Gupta *et al.*, 2021; Pirson *et al.*, 2023). Cooling those systems and generating the electricity that feeds them consumes freshwater at scales large enough to matter regionally (Mytton, 2021). The semiconductor supply chain draws on scarce materials and discharges hazardous waste (Sandhu *et al.*, 2025), while the shortening replacement cycles that keep systems competitive turn still-functional hardware into electronic waste. Across its full lifecycle, then, an HPC system carries costs in carbon, water, and materials and leaves electronic waste behind, well beyond the operational energy it draws.

Visibility into this broader footprint has only recently begun to grow, and the established ways of evaluating and operating HPC systems still capture only part of it. The Green500 ranks systems by performance per watt under a single benchmark (Feng and Cameron, 2007), reducing sustainability to an efficiency ratio measured on Linpack. Job schedulers decide when and where computation runs, yet they have conventionally been designed to prioritize performance above all else (Kocot *et al.*, 2023; Czarnul *et al.*, 2019), leaving environmental cost outside the optimization objective. Procurement and hardware-replacement decisions weigh performance and acquisition cost while treating lifecycle impact as an afterthought (Li *et al.*, 2023). Each of these approaches sharpens a single dimension of a problem whose full extent remains only partially charted, and the gains they deliver are repeatedly absorbed by growing demand, so the best-measured dimension does not improve in absolute terms (York *et al.*, 2022). Attention to the dimensions these approaches leave out is a comparatively recent and still-emerging area of interest.

This work begins with a systematic mapping study that turns this emerging interest into a concrete gap (Rosa *et al.*, 2026). Classifying 62 reports from sixteen years of research across environmental impacts, lifecycle stages, system loci, and management levers, the study finds the literature concentrated in a single corridor of that classification: carbon among the impacts, operations among the lifecycle stages, facility infrastructure among the system loci, and measurement among the management levers. The dimensions outside that corridor sit in sparsely populated cells, studied rarely and in isolation: water beyond cooling, materials beyond what carbon accounting already captures, biodiversity, and end-of-life processing. The rest of this work sets out to fill the gaps the map identifies, and it currently runs along two branches.

The first branch focuses on HPC systems' operation. The batch scheduler decides when and where jobs run on performance grounds, users submit work without seeing its environmental cost, and hardware-replacement decisions weigh performance and price while overlooking embodied and end-of-life impacts. This work responds by building simulation infrastructure to test a range of mechanisms: scheduling heuristics that react to a time-varying grid, user-facing accounting that makes cost visible, platform controls that trade performance for a smaller footprint, and lifecycle-aware policies that tie hardware aging and replacement to an environmental budget. The infrastructure and its footprint model track operational carbon, water, and embodied impact, so this work can study these mechanisms without access to a production system. A first result along this line is Greenfilling, a small change to EASY backfilling (Skovira *et al.*, 1996) that holds back opportunistic job starts when grid carbon intensity runs above its recent average. Evaluated across three national grids and two production machines, it bought certain delay for uncertain environmental return, which sets a lower bound on how sophisticated such an intervention has to be.

The second branch focuses on HPC systems' impacts as a whole. The Green500 ranks systems by performance per watt, but this is just one side of the real impacts of using those platforms. Two systems can rank side by side while drawing very different absolute power, sitting on grids whose water and carbon intensity differ by orders of magnitude, carrying different embodied impacts, and consuming water at rates that do not match the regional scarcity. The aim is not to replace the Green500 but to use it as a reference point, showing what a sufficiency-aware lens reveals: cases where efficiency improves while absolute burden grows, where embodied costs dominate but stay invisible, and where regional context changes which system is actually less harmful. A first result along this line is an initial literature analysis that identified four candidate layers for such a lens and mapped the evidence coverage across them.

The remainder of this document is organized as follows. Chapter 2 reviews the environmental footprint of HPC across its lifecycle. Chapter 3 presents the systematic mapping study and the classification behind the gap this work addresses. Chapter 4 reports the preliminary results on both branches. Chapter 5 sets out what remains. The first branch adds user modeling, platform controls, and lifecycle-aware policies to the scheduling mechanism tested so far. The second develops a sufficiency-aware framework and builds an empirical case, from modeled and real systems, for asking whether and how much additional computation is environmentally justified. Chapter 6 closes with the schedule and the expected contributions of the completed work.

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Chapter 3

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5.1 Accounting for Environmental Cost in HPC Operations

5.2 Sufficiency-Aware Evaluation of HPC Systems

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Schedule and Expected Contributions

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